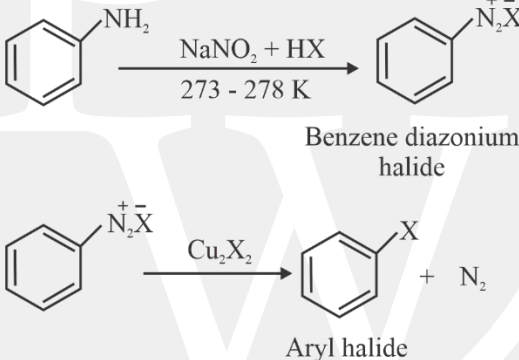
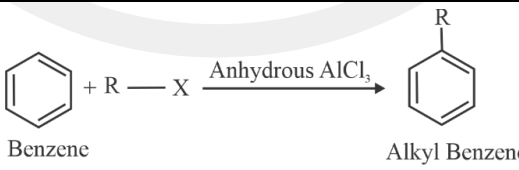
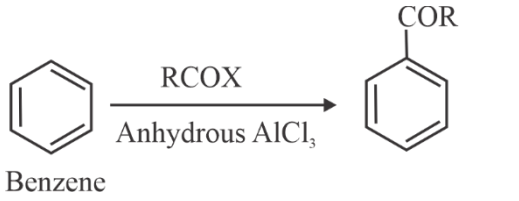
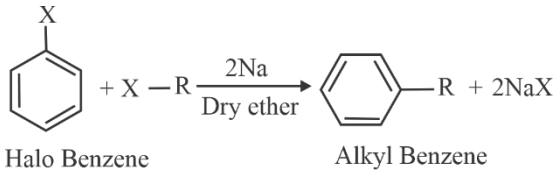
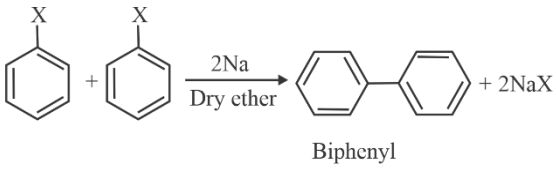
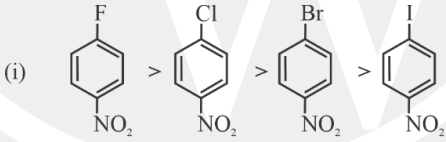
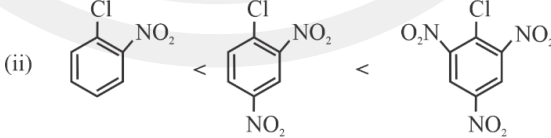
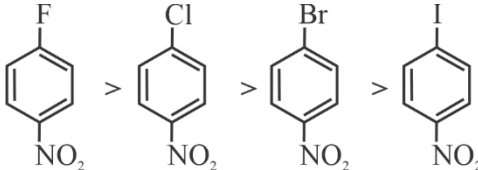
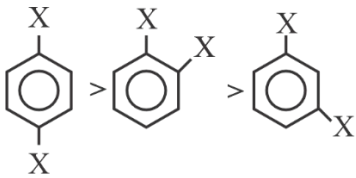


Haloalkanes and Haloarenes

Sr. No.	Concept	Description	Other information
1.	Name reactions		
	(a) Finkelstein reaction	$RX + NaI \longrightarrow R-I + NaX$	Reagent = Acetone X = Br, Cl
	(b) Swarts reaction	$RX + AgF \longrightarrow RF + X$	X = Br, Cl
	(c) Sandmeyer's reaction	 $\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow[273-278\text{ K}]{\text{NaNO}_2 + \text{HX}} \text{C}_6\text{H}_5\text{N}_2^+\text{X}^-$ Benzene diazonium halide $\text{C}_6\text{H}_5\text{N}_2^+\text{X}^- \xrightarrow{\text{Cu}_2\text{X}_2} \text{C}_6\text{H}_5\text{X} + \text{N}_2$ Aryl halide	Reagent = Cu_2X_2 X = Cl, Br
	(d) Friedel Craft's reaction		
	(i) Friedel Craft Alkylation	 $\text{C}_6\text{H}_6 + \text{R-X} \xrightarrow{\text{Anhydrous AlCl}_3} \text{C}_6\text{H}_5\text{R}$ Benzene Alkyl Benzene	Reagent = Anhydrous AlCl_3 X = Cl, Br, I
	(ii) Friedel Craft Acylation	 $\text{C}_6\text{H}_6 \xrightarrow[\text{Anhydrous AlCl}_3]{\text{RCOX}} \text{C}_6\text{H}_5\text{COR}$ Benzene	Reagent = Anhydrous AlCl_3 X = Cl, Br, I
	(e) Wurtz reaction	$2RX + 2Na \longrightarrow R-R + 2NaX$	X = Br, Cl, Reagent = Na, dry ether

Sr. No.	Concept	Description	Other information
	(f) Wurtz-fittig reaction	 Halo Benzene Alkyl Benzene	$X = \text{Cl, Br, I}$ Reagent = Na, Dry ether
	(g) Fittig reaction	 Biphenyl	$X = \text{Cl, Br}$ Reagent = Na, Dry ether
2.	Reactivity order of alkyl halides towards S_N1 Nucleophilic substitution reaction	$3^\circ > 2^\circ > 1^\circ > -\text{CH}_3$	Reactivity $\propto$ Stability of carbocation
3.	Reactivity order of alkyl halides towards S_N2 Nucleophilic substitution reaction	$-\text{CH}_3 > 1^\circ > 2^\circ > 3^\circ$	Reactivity $\propto \frac{1}{\text{Steric hindrance}}$
4.	Physical properties of Haloalkanes		
	(a) Dipole moment of RX	$\text{RCl} > \text{RF} > \text{RBr} > \text{RI}$	$\text{R} = \text{CH}_3$ Dipole moment decreases with decrease in electronegativity of halogen.
	(b) Bond enthalpies	$\text{RF} > \text{RCl} > \text{RBr} > \text{RI}$	The size of halogen atom increases from F to I hence bond length from R-F to R-I increases due to decrease of effective nuclear charges on electrons. $\therefore$ Bond enthalpy from R - F to R - I decreases.

Sr. No.	Concept	Description	Other information
	(c) Boiling point	$RI > RBr > RCl > RF$	This is because of increase in size and mass of halogen atoms. $B.P \propto \frac{1}{\text{Branching}}$
	(d) Density	$R - I > R - Br > R - Cl$	Density of alkyl halide $\propto$ Number of carbons atoms of alkyl halide $\propto$ Number of halogens atoms of alkyl halide $\propto$ Atomic weight of halogen atoms.
5.	Reactivity of Haloalkanes Towards Nucleophile	$RI > RBr > RCl > RF$	The reactivity of the given haloalkanes towards nucleophile depend on leaving group ability of halogens. The leaving group ability of halogens are in the order $I^- > Br^- > Cl^- > F^-$.
6.	Reactivity of Haloarenes Towards Nucleophile	(i)  (ii) 	With increase in number of strong electron withdrawing groups at ortho and para position, reactivity of haloarene towards nucleophilic increases.
7.	Physical Properties of Haloarenes		
	(a) Boiling point		Boiling points depends on mass.

Sr. No.	Concept	Description	Other information
	(b) Melting point		X = Cl, Br, I The para-isomer has higher melting point than the ortho and meta-isomers, this is due to symmetrical structure of the para isomer due to which it is better packed in the crystal lattice.
7.	Grignard reagent	$\text{R} - \text{X} + \text{Mg} \xrightarrow{\text{dry ether}} \underset{\text{Grignard reagent}}{\text{R} - \text{MgX}}$	X = Cl, Br, I Reagent = Dry ether

